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**Course Name:**5G 6G Communication Systems

**Course Code:**ECA5601

**Ex:11 Implementation of IoT-based Smart Agriculture using MATLAB Date**

**Aim:**

To design and simulate an IoT-based smart agriculture system using MATLAB by monitoring soil moisture levels, controlling water pump status (ON/OFF), and evaluating irrigation efficiency for optimized water resource management.

**Objective: Analyze Channel Capacity (bps/Hz)**

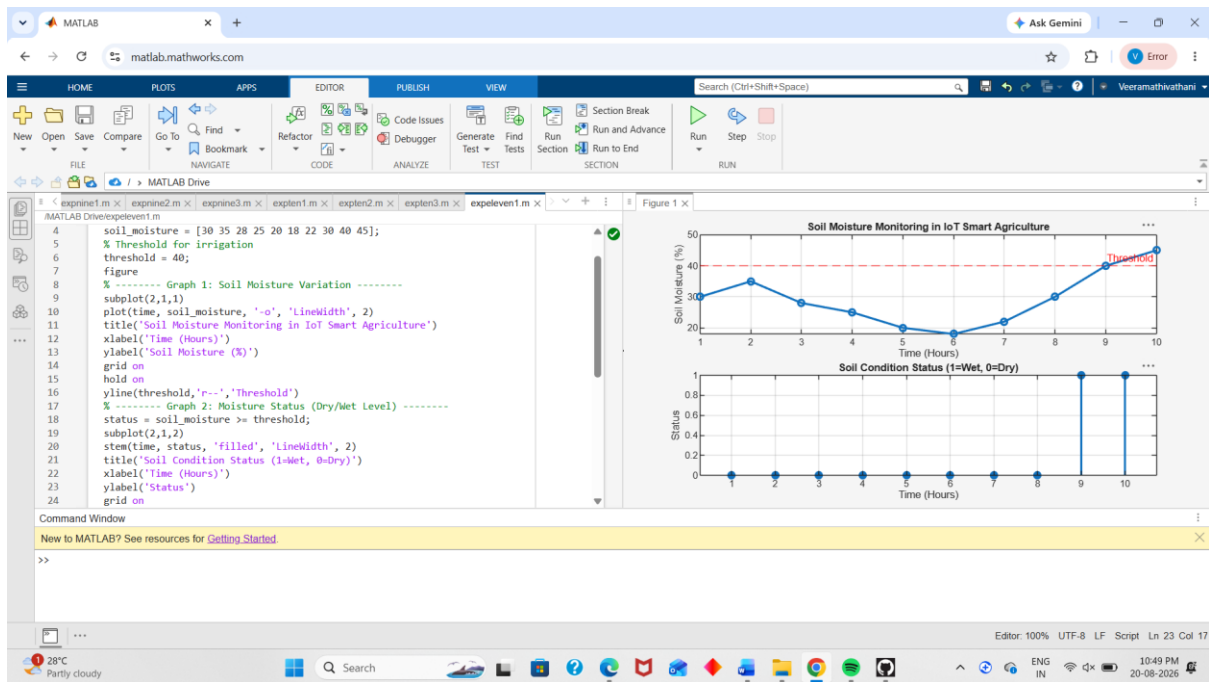
- 1.To monitor the Soil Moisture (%) level in agricultural fields using IoT-based sensing simulation in MATLAB.
- 2.To control and display the Water Pump Status (ON/OFF) based on threshold soil moisture conditions.
- 3.To evaluate the Irrigation Efficiency (%) by analyzing water usage effectiveness and moisture maintenance level.

**Procedure**

1. Define soil moisture range (0–100%) and set an irrigation threshold value (e.g., 40%).
- 2.Generate or input soil moisture data to simulate field conditions.
- 3.Compare soil moisture with the threshold value to determine pump status (ON if low moisture, OFF if sufficient moisture).
- 4.Simulate water pumping action based on control logic.
- 5.Calculate irrigation efficiency based on moisture improvement and water usage.
- 6.Display soil moisture level, pump status, and efficiency results.
7. Plot graphs to analyze soil moisture variation and irrigation performance. 8. Evaluate system behavior for smart agriculture decision-making.

## Objective 1: Matlab Code and Output

```
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold for irrigation
threshold = 40;
figure
% ----- Graph 1: Soil Moisture Variation -----
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2)
title('Soil Moisture Monitoring in IoT Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')
% ----- Graph 2: Moisture Status (Dry/Wet Level) -----
status = soil_moisture >= threshold;
subplot(2,1,2)
stem(time, status, 'filled', 'LineWidth', 2)
title('Soil Condition Status (1=Wet, 0=Dry)')
xlabel('Time (Hours)')
ylabel('Status')
grid on
```



## Objective 2: Matlab Code and Output

clc; clear; close all;

% Simulated soil moisture (%)

time = 1:10;

soil\_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level

threshold = 40;

% Pump control logic (1 = ON, 0 = OFF)

pump\_status = soil\_moisture < threshold;

figure

% ----- Graph 1: Soil Moisture vs Pump Status -----

subplot(2,1,1)

plot(time, soil\_moisture, '-o','LineWidth',2)

title('Soil Moisture vs Pump Control')

xlabel('Time (Hours)')

ylabel('Soil Moisture (%)')

grid on

hold on

```
ylines(threshold,'r--','Threshold')
```

% ----- Graph 2: Pump Status (ON/OFF) -----

```
subplot(2,1,2)
```

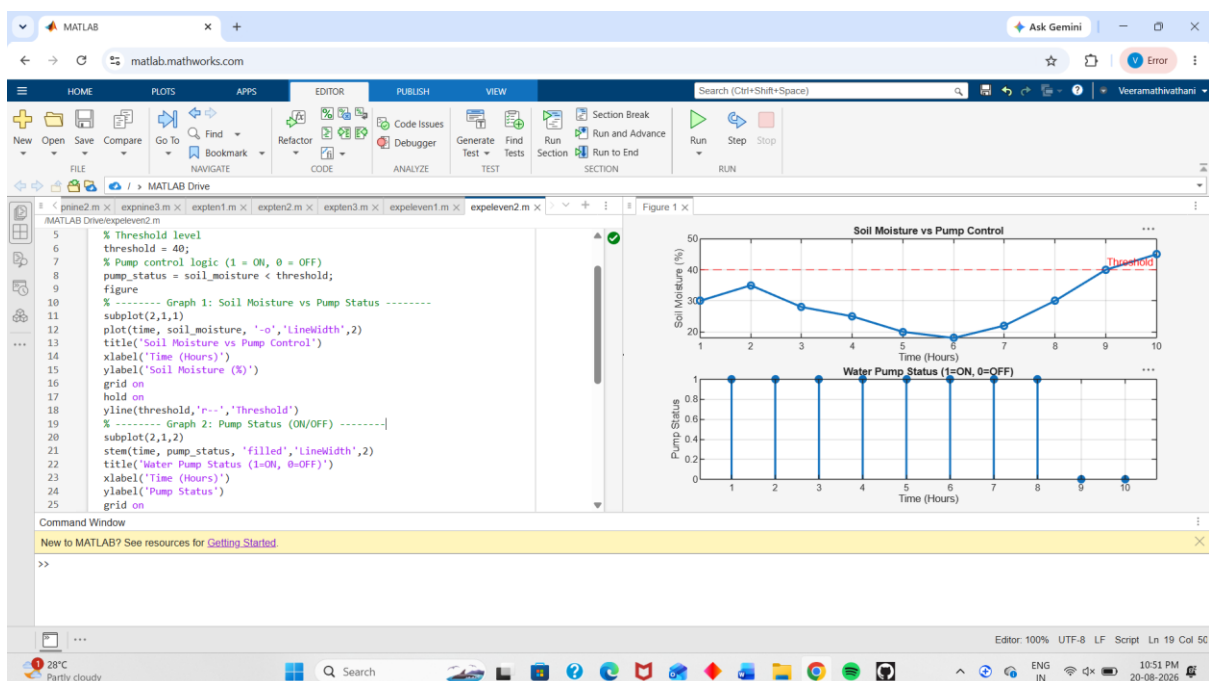
```
stem(time, pump_status, 'filled','LineWidth',2)
```

```
title('Water Pump Status (1=ON, 0=OFF)')
```

```
xlabel('Time (Hours)')
```

```
ylabel('Pump Status')
```

grid on



### Objective 3: Matlab Code and Output

```
clc; clear; close all;
```

% Simulated soil moisture data (%)

```
time = 1:10;
```

```
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
```

% Threshold level

```
threshold = 40;
```

% Irrigation efficiency calculation (%)

```
efficiency = (soil_moisture ./ threshold) * 100;
```

```
efficiency(efficiency > 100) = 100; % limit to 100%
```

figure

```
% ----- Graph 1: Soil Moisture vs Efficiency -----
```

```
subplot(2,1,1)
```

```
plot(time, soil_moisture, '-o','LineWidth',2)
```

```
title('Soil Moisture Variation')
```

```
xlabel('Time (Hours)')
```

```
ylabel('Soil Moisture (%)')
```

```
grid on
```

```
hold on
```

```
ylines(threshold,'r--','Threshold')
```

```
% ----- Graph 2: Irrigation Efficiency -----
```

```
subplot(2,1,2)
```

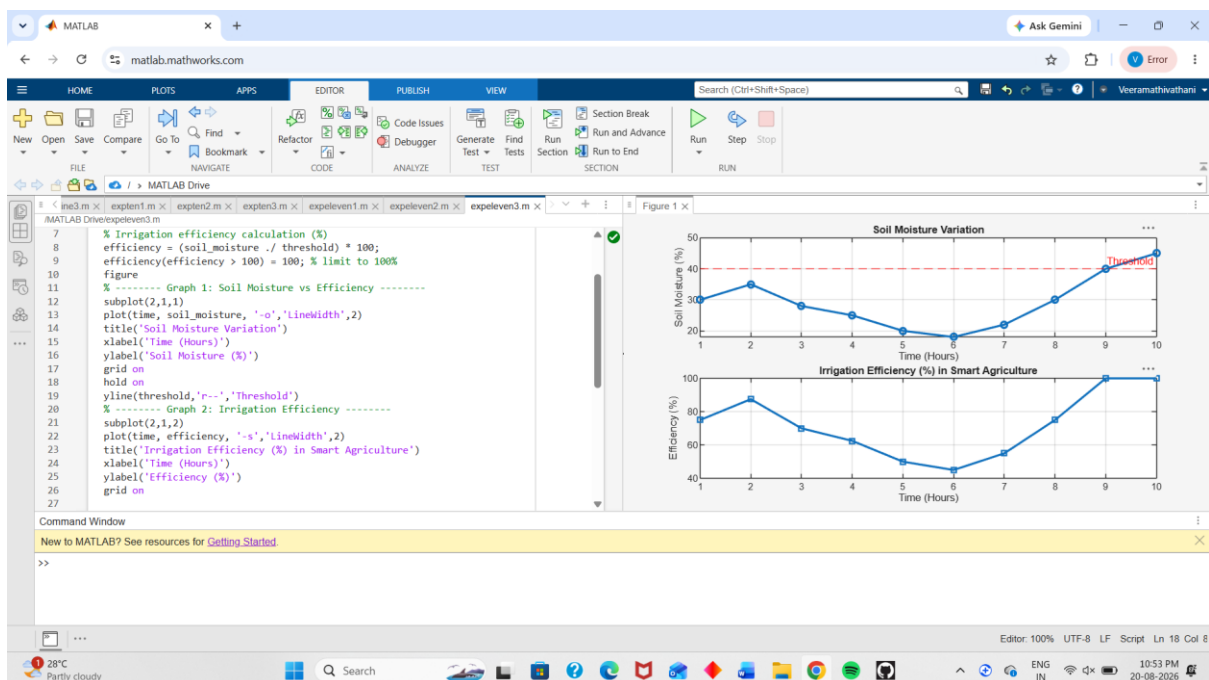
```
plot(time, efficiency, '-s','LineWidth',2)
```

```
title('Irrigation Efficiency (%) in Smart Agriculture')
```

```
xlabel('Time (Hours)')
```

```
ylabel('Efficiency (%)')
```

```
grid on
```



**Result:**

1. The soil moisture levels were successfully monitored using MATLAB simulation, showing variation over time and identifying dry and wet conditions based on threshold values.
2. The water pump control system was effectively implemented, where the pump automatically switched ON when soil moisture dropped below the threshold and remained OFF when moisture was sufficient.
3. The irrigation efficiency was successfully calculated and analyzed, showing that the system optimizes water usage by maintaining soil moisture close to the desired level